IB DP CHEMISTRY

Complete Cheat Sheet

Full Syllabus . HL + SL . Score a 7

Stoichiometry | Atomic Structure | Periodicity | Bonding | Energetics
Kinetics | Equilibrium | Acids & Bases | Redox | Organic Chem

20 Pages | All Formulas | Exam Tips | Visual Diagrams | HL Extensions

Designed for IB DP students aiming for the highest marks

How to Use This Cheat Sheet

Purpose

* Fast-revision reference for IB DP Chemistry (SL + HL)
* Each section = one exam topic with formulas, diagrams & tips
* HL content clearly labeled with **[HL] HL** tags

Reading Order

* **First pass:** Read the section headers + formula boxes only
* **Second pass:** Study the exam tips and common mistakes
* **Final review:** Use Pages 17-20 for ultra-condensed revision

Icons & Boxes

Icon	Meaning
[TIP] EXAM TIP	High-value exam strategy
[!] MISTAKE	Common error to avoid
[>] INSIGHT	Examiner perspective
[HL] HL	Higher Level only content
[F] FORMULA	Key equation in blue box

Syllabus Roadmap

#	Topic	SL Pages	HL Extra	Key Skill
1	Stoichiometry	3-4	--	Mole calculations, gas laws
2	Atomic Structure	5	5	Electron config, IE trends
3	Periodicity	6	6	Trend graphs, explanations
4	Bonding	7-8	8	VSEPR, IMFs, polarity
5	Energetics	9	9	Hess's Law, Born-Haber (HL)
6	Kinetics	10	10	Rate laws, Ea, catalysis
7	Equilibrium	11	11	Kc, Le Chatelier, Kp (HL)
8	Acids & Bases	12	12	pH, Ka, titration curves
9	Redox	13	13	Half-equations, cells
10	Organic Chem	14-16	14-16	Mechanisms, pathways

[TIP] EXAM TIP
Start with your weakest topic. Use the formula boxes as flashcards -- cover the labels and try to recall units and context.

1 STOICHIOMETRY

Moles . Gas Laws . Solutions . Yield

The Mole Concept

* 1 mol = 6.022 x 1023 particles (Avogadro's number, N_A)
* Molar mass *M* = mass per mole (g mol-1) -- from periodic table
* Relative atomic mass = weighted avg of isotope masses

Empirical vs Molecular Formula

* **Empirical formula:** simplest whole-number ratio of atoms
* **Molecular formula:** actual number of atoms in one molecule
* Molecular formula = n x empirical formula; find n = M_{mol}/M_{emp}

Limiting Reagent

* The reactant that is **completely consumed** first
* Determines the maximum theoretical yield
* Method: divide moles of each reactant by its stoichiometric coefficient; smallest ratio = limiting reagent

Concentration & Dilution

* c = n / V (mol dm-3, also written M)
* Dilution: c1V1 = c2V2
* V must be in dm3 (= litres); 1 dm3 = 1000 cm3

Key Formulas

n = m / M

n=moles | m=mass(g) | M=molar mass(g/mol)

n = c x V

n=moles | c=conc(mol/dm3) | V=vol(dm3)

PV = nRT

P(Pa).V(m3)=n(mol).R.T(K) | R=8.314 J/mol.K

% yield = actual/theoretical x 100

actual & theoretical in same units

urity = mass pure/mass sample x

Gas Law Constants

* R = 8.314 J mol-1 K-1
* At STP (0 CC, 100 kPa): 1 mol gas = 22.7 dm3
* T(K) = T(degC) + 273 <- always convert!

[!] COMMON MISTAKE

Always convert cm3 -> dm3 (/1000) and degC -> K (+273) before using PV=nRT. Forgetting unit conversions is the #1 stoichiometry error.

Worked Calculation -- Limiting Reagent

Step	Action	Example: 2H2 + O2 -> 2H2O (given 4g H2, 32g O2)
1	Find moles of each reactant	n(H2)=4/2=2 mol; n(O2)=32/32=1 mol
2	Divide by stoichiometric coeff.	H2: 2/2=1; O2: 1/1=1
3	Smallest ratio = limiting reagent	Both equal -> neither is limiting here
4	Use LR to find theoretical yield	n(H2O)=2 mol -> mass=2x18=36 g

[>] EXAM INSIGHT

In % yield questions, identify the limiting reagent first. Theoretical yield always comes from the limiting reagent -- never use excess reagent moles.

Atom Economy (HL + Green Chemistry)

Atom Economy = M(desired) / SM(all products) x 100

Measures efficiency of a reaction -- high = greener

* High atom economy = less waste, more sustainable process
* Addition reactions have 100% atom economy; substitution reactions are lower

[HL] HL EXTENSION

Atom economy is a green chemistry concept. Link it to industrial processes in paper 2 extended-response questions.

2 ATOMIC STRUCTURE

Orbitals . Electron Config . Ionization Energy

Subatomic Particles

Particle	Mass (u)	Charge	Location
Proton	1	+1	Nucleus
Neutron	1	0	Nucleus
Electron	~1/1836	-1	Shells/orbitals

Orbitals & Subshells

* s orbital: spherical, 1 orbital, max 2e-
* p orbital: dumbbell-shaped, 3 orbitals, max 6e-
* d orbital: complex, 5 orbitals, max 10e- [HL] HL
* f orbital: 7 orbitals, max 14e- [HL] HL

* **Aufbau principle:** fill lowest energy first
* **Pauli exclusion:** max 2e- per orbital, opposite spins
* **Hund's rule:** one e- in each orbital before pairing

Electron Configuration Examples

* Na (Z=11): 1s2 2s2 2p6 3s1 or [Ne] 3s1
* Fe (Z=26): [Ar] 3d6 4s2 [HL] HL
* Cu (Z=29): [Ar] 3d10 4s1 (exception!) [HL] HL
* Cr (Z=24): [Ar] 3d5 4s1 (exception!) [HL] HL

Ionization Energy

* **First IE:** energy to remove 1 mol e- from gaseous atoms
* X(g) -> X+(g) + e-; DH = +IE
* Successive IEs always increase (harder to remove from +ve ion)
* Large jump in successive IE -> indicates electron removed from inner shell

IE Trends

Trend	Direction	Reason
Across period	(up) increases	More protons, same shielding -> stronger nuclear attraction
Down group	(down) decreases	More shells -> greater shielding + increased atomic radius
Exceptions	Al < Mg, S < P	Al: 3p more shielded than 3s; S: paired e- repulsion

Isotopes & Mass Spectrometry

* Isotopes: same Z (protons), different A (neutrons)
* Relative atomic mass = S(isotope mass x abundance)
* Mass spec: ionization -> acceleration -> deflection -> detection
* Highest m/z peak with loss of 1 = M+ peak (molecular ion) [HL] HL

[>] EXAM INSIGHT

When asked for electron config of ions, remove e- from highest n first (e.g., F 3d).

[HL] HL EXTENSION

Exceptions: Cu and Cr have unusual configs due to extra stability of half-filled/fully-filled d subshells. These are regularly tested at HL.

3

PERIODICITY

Trends . Properties . Explanations

Key Periodic Trends

Property	Across Period (->)	Down Group ((down))	Reason
Atomic radius	(down) decreases	(up) increases	More protons (across); more shells (down)
Ionic radius	Cations < atom; anions > atom	(up) increases	Electron count changes; more shells down
Electronegativity	(up) increases (F highest)	(down) decreases	Stronger nuclear pull on bonding e- across period
Melting point	Varies by structure	Varies	Depends on bonding type, not simple trend
1st Ionization E.	(up) general increase	(down) decreases	Nuclear charge (across); shielding/distance (down)
Metallic character	(down) decreases	(up) increases	Metals lose e- easily; more shells = weaker hold

Period 3 Elements (Na->Ar)

* **Na, Mg:** metallic bonding -> good conductors, low-medium MP
* **Al:** metallic lattice, high MP (3D metallic bonds)
* **Si:** giant covalent network -> very high MP
* **P, S, Cl, Ar:** simple molecular -> low MP (weak London forces)
* **S** has higher MP than P/Cl due to larger S8 molecules

Period 3 Oxides

* Na2O, MgO: basic (react with acid -> salt + H2O)
* Al2O3: amphoteric (reacts with acid AND base)
* SiO2, P4O10, SO3, Cl2O7: acidic (react with base -> salt + H2O)

Period 3 Chlorides

* NaCl, MgCl2: ionic -> dissolve in water, pH ~7 (NaCl) or slightly acidic (MgCl2)
* AlCl3: ionic with covalent character -> Al3+ hydrolyzes -> acidic
* SiCl4, PCl3, PCl5: covalent -> react violently with water -> HCl gas

Electronegativity & Bonding

* DEN > 1.7: ionic bond (rough guide)
* DEN 0.4-1.7: polar covalent bond
* DEN < 0.4: nonpolar covalent bond
* Pauling scale: F=4.0, O=3.5, N=3.0, Cl=3.2

[>] EXAM INSIGHT

Always explain trends using: (1) nuclear charge, (2) shielding, (3) distance from n

for full marks.

[TIP] EXAM TIP

The exceptions in IE across Period 2 (B < Be, O < N) are common exam questions. B: 2p is higher energy than 2s. O: paired 2p e- repel each other.

4 BONDING

Ionic . Covalent . Metallic . IMFs . VSEPR

Types of Chemical Bonds

Bond Type	Formed Between	Transfer/Share?	Properties
Ionic	Metal + non-metal	Transfer of e-	High MP, brittle, conducts when molten/dissolved
Covalent	Non-metal + non-metal	Sharing of e-	Varies by structure (simple vs giant)
Metallic	Metal atoms	Delocalized e- sea	Conducts, malleable, high MP (esp. transition metals)
Coordinate	Lone pair donor + acceptor	Sharing (dative)	Same as covalent once formed; labeled with arrow

Lewis (Dot-and-Cross) Structures

* Draw all valence electrons
* Each bond = 2 shared electrons
* Lone pairs satisfy octet rule for each atom
* Exceptions: BF3 (6e- on B), PCl5/SF6 (expanded octet) [HL] HL

Bond Properties

Bond Order	Length	Strength	Example
Single (sigma)	Longest	Weakest	C-C in ethane
Double (sigma+pi)	Medium	Medium	C=C in ethene
Triple (sigma+2pi)	Shortest	Strongest	C=C in ethyne

Resonance Structures [HL] HL

* Some molecules cannot be described by a single Lewis structure
* E.g., O3, SO2, benzene (C6H6), CO32-, NO3-
* Bond order = average over resonance structures
* Benzene: bond order = 1.5 (between single and double)

Intermolecular Forces (IMFs)

IMF	Strength	Present When?
London dispersion (van der Waals)	Weakest	ALL molecules; (up) with molar mass
Dipole-dipole	Moderate	Polar molecules (permanent dipole)
Hydrogen bond	Strongest IMF	N-H, O-H, or F-H present

* **H-bond:** lone pair donor (O, N, F) + H bonded to O/N/F
* H-bonds explain anomalous high BP of H2O, HF, NH3
* London forces dominate in large, nonpolar molecules (e.g., I2)

Polarity & Molecular Dipole

* Polar bond: unequal sharing (DEN > 0)
* Polar molecule: polar bonds + **asymmetric** shape
* CO2: polar bonds, linear -> symmetric -> nonpolar molecule
* H2O: polar bonds, bent -> asymmetric -> polar molecule

[>] EXAM INSIGHT

CCl4 has polar bonds but is nonpolar overall (tetrahedral, dipoles cancel). CH2Cl2 is polar (asymmetric). Know the difference!

VSEPR Shapes & Molecular Geometry

Formula	Bond Pairs	Lone Pairs	Shape	Angle	Example
AX2	2	0	Linear	180 deg	BeCl2, CO2
AX3	3	0	Trigonal planar	120 deg	BF3, SO3
AX2E	2	1	Bent/V-shaped	<120 deg	SO2
AX4	4	0	Tetrahedral	109.5 deg	CH4, CCl4
AX3E	3	1	Trigonal pyramidal	<109.5 deg	NH3
AX2E2	2	2	Bent/V-shaped	<109.5 deg	H2O (104.5 deg)
AX5	5	0	Trigonal bipyramidal	90 deg/120 deg	PCl5 [HL] HL
AX6	6	0	Octahedral	90 deg	SF6 [HL] HL

[TIP] EXAM TIP

Lone pairs repel MORE than bonding pairs. Each lone pair reduces bond angle by ~2.5 deg. So NH3 ~107 deg, H2O ~104.5 deg.

Shape Diagrams



180 deg

Linear 180 deg



120 deg

Trigonal Planar 120 deg



109.5 deg

Tetrahedral 109.5 deg



<109.5 deg

Bent <109.5 deg



<109.5 deg

Trig. Pyr. <109.5 deg



90 deg

Octahedral 90 deg

5 ENERGETICS

Enthalpy . Hess's Law . Bond Energies . Thermodynamics

Key Formulas

q = mcDT

q=heat(J) | m=mass(g) | c=4.18 J/g.K | DT=temp change

DH = -q/n

DH in kJ/mol | n=moles of limiting reagent

h = SDHfdeg(products) - SDHfdeg(r

Standard enthalpies of formation

ldegrxn = SBE(broken) - SBE(forme

Bond enthalpies (always endothermic to break)

Enthalpy Diagrams

* **Exothermic:** DH < 0; products lower than reactants; releases heat
* **Endothermic:** DH > 0; products higher than reactants; absorbs heat
* **Activation energy (Ea):** energy barrier -- height from reactants to transition state

Hess's Law

* DH of a reaction is independent of the pathway taken
* If equation reversed, sign of DH changes
* If equation multiplied by n, DH multiplied by n
* Use enthalpy cycles: arrange equations so intermediates cancel

Calorimetry Notes

* Use c = 4.18 J g-1 K-1 for aqueous solutions
* Assume density of solution = 1 g cm-3
* DH is negative for exothermic reactions (T of solution rises)
* Measured DH < theoretical -> heat lost to surroundings

Standard Conditions

* 100 kPa, 298 K (25 degC), 1 mol dm-3 for solutions
* DHfdeg(element in standard state) = 0 by definition

[HL] HL: Born-Haber Cycles

* For ionic compounds: construct energy cycle using:
* DHfdeg = DHat(metal) + DHat(non-metal) + IE + EA + DHlattice
* Lattice enthalpy: energy to form gaseous ions from lattice
* More exothermic lattice DH -> more stable ionic compound
* High charge + small ions -> large lattice enthalpy

[HL] HL: Entropy & Gibbs Energy

DG = DH - TDS

DG<0: spontaneous | T in Kelvin | DS in J/mol.K

* DS > 0: disorder increases (gases, dissolution, more particles)
* Spontaneous when DG < 0

[HL] HL EXTENSION

DG < 0 means spontaneous, not necessarily fast. Kinetics (Ea) determines spe

determines direction.

[!] COMMON MISTAKE

In calorimetry, the DH you calculate is for the number of moles you used -- always divide by moles of

the reaction to get kJ/mol.

6 KINETICS

Rate . Collision Theory . Activation Energy . Catalysis

Rate of Reaction

* Rate = change in concentration per unit time
* Units: mol dm-3 s-1
* Measured by: change in concentration, gas volume, mass loss, light absorption

Factors Affecting Rate

Factor	Effect on Rate	Why?
(up) Temperature	(up) increases	More particles exceed Ea; more frequent + energetic collisions
(up) Concentration	(up) increases	More particles per volume -> more frequent collisions
(up) Surface area	(up) increases	More particles exposed -> more collisions possible
Catalyst	(up) increases	Provides alternative pathway with lower Ea
(up) Pressure (g)	(up) increases	Higher concentration of gas particles

Collision Theory

* Reaction occurs only when particles collide with:
* (1) Sufficient energy >= activation energy (Ea)
* (2) Correct orientation of colliding particles

Maxwell-Boltzmann Distribution

* Graph: molecular speed/energy vs fraction of molecules
* Area under curve = total number of molecules (constant)
* Effect of temperature: curve flattens, shifts right, peak moves right
* Effect of catalyst: Ea moves left on same distribution -> more molecules exceed Ea

[HL] HL: Rate Equations

Rate = k[A]m[B]n

k=rate constant | m,n=reaction orders (found experimentally)

* **Zero order:** rate independent of [A]; graph [A] vs t is linear
* **First order:** rate proportional to [A]; graph ln[A] vs t is linear
* **Second order:** rate proportional to [A]2; graph 1/[A] vs t is linear
* **Overall order** = m + n

Determining Order (HL)

* Compare experiments: if [A] doubles and rate doubles -> 1st order in A
* If [A] doubles and rate quadruples -> 2nd order in A
* If [A] doubles and rate unchanged -> zero order in A

Arrhenius Equation (HL)

k = Ae^(-Ea/RT)

A=frequency factor | Ea=activation energy | R=8.314

ln k = ln A - Ea/RT

Plot ln k vs 1/T -> slope = -Ea/R

Catalysis

* **Homogeneous catalyst:** same phase as reactants (e.g., H+ in esterification)
* **Heterogeneous catalyst:** different phase (e.g., Fe in Haber process, V2O5 in Contact process)
* Catalyst increases rate by lowering Ea -- NOT shifting equilibrium

[TIP] EXAM TIP

Catalyst lowers Ea for both forward AND reverse reactions equally. This spee

equilibrium, not shifting it.

[!] COMMON MISTAKE

Reaction orders cannot be determined from stoichiometric coefficients -- only from experimental rate data. This is a very common exam mistake.

7 EQUILIBRIUM

Kc . Kp . Le Chatelier's Principle . Haber Process

Dynamic Equilibrium

* Forward and reverse rates are equal
* Concentrations remain constant (not equal)
* Only reached in closed system

Equilibrium Constant Kc

* For: aA + bB <-> cC + dD

Kc = [C]c[D]d / [A]a[B]b

Square brackets = equilibrium concentrations (mol/dm3)

* **Kc > 1:** equilibrium favors products (right)
* **Kc < 1:** equilibrium favors reactants (left)
* **Kc = 1:** balanced mixture
* Kc only changes with **temperature**
* Pure solids and liquids NOT included in Kc expression

Le Chatelier's Principle

* System opposes any imposed change to restore equilibrium

Change	System Response
(up) Concentration of reactant	Shifts RIGHT (->) to consume it
(up) Concentration of product	Shifts LEFT (<-) to consume it
(up) Temperature (exothermic rxn)	Shifts LEFT (<-); Kc decreases
(up) Temperature (endothermic rxn)	Shifts RIGHT (->); Kc increases
(up) Pressure (on gas eq.)	Shifts to side with fewer gas moles
Add catalyst	No shift -- reaches eq. faster only

[HL] HL: Equilibrium Constant Kp

Kp = p(C)c x p(D)d / p(A)a x p(B)b

p = partial pressure (kPa or Pa)

* Partial pressure: p = mole fraction x total pressure
* Kp = Kc(RT)^Dn where Dn = moles gas products - reactants [HL] HL

Reaction Quotient Qc [HL] HL

* Qc = same expression as Kc but using any concentrations
* Qc < Kc -> system shifts right to reach eq.
* Qc > Kc -> system shifts left to reach eq.
* Qc = Kc -> at equilibrium

Industrial Equilibria

* **Haber Process (N2 + H2 -> NH3):**
* Conditions: 400-500 degC, 200 atm, Fe catalyst
* High pressure favors products (4 mol -> 2 mol gas)
* Low T favors products (exothermic) BUT too slow
* Compromise: 400-500 degC for acceptable rate + yield
* **Contact Process (SO2 + O2 -> SO3):**
* Conditions: 450 degC, 1-2 atm, V2O5 catalyst

[>] EXAM INSIGHT

Catalyst is added in the Haber process for rate only. It does NOT impro

common misconception.

[!] COMMON MISTAKE

Always write Kc without units (pure number). Remember: only temperature changes Kc. Pressure, concentration, and catalysts do not change Kc.

8 ACIDS & BASES

pH . Ka . Buffers . Titration Curves

Definitions

Theory	Acid	Base
Arrhenius	Produces H+ in water	Produces OH- in
Bronsted-Lowry	Proton (H+) donor	Proton (H+) acceptor
Lewis [HL] HL	Electron pair acceptor	Electron pair donor

Key Formulas

pH = -log[H+]

[H+] in mol/dm3

[H+] = 10^(-pH)

Reverse calculation

= [H+][OH-] = 1x10-14 (at 25 deg

pKw = 14; pH + pOH = 14

Ka = [H+][A-] / [HA]

Weak acid equilibrium expression

pH = 1/2(pKa - log c)

Weak acid pH approximation

Strong vs Weak Acids/Bases

* **Strong:** fully dissociated; [H+] = initial concentration (pH = -log[H+])
* **Weak:** partially dissociated; Ka expression needed; higher pH than strong acid at same conc.
* pKa = -log Ka; smaller pKa = stronger acid

Titration Curves

Titration Type	Equiv. Point pH	Indicator
Strong acid + strong base	~7	Any (phenolphthalein or methyl orange)
Weak acid + strong base	>7 (~8-9)	Phenolphthalein (pH 8.2-10)
Strong acid + weak base	<7 (~5-6)	Methyl orange (pH 3.1-4.4)
Weak acid + weak base	~7 (variable)	No sharp endpoint -- difficult

Buffer Solutions [HL] HL

* Resist pH change on addition of small amounts of acid or base
* **Weak acid buffer:** weak acid + its conjugate base salt
* E.g., CH3COOH + CH3COONa
* **Weak base buffer:** weak base + its conjugate acid salt
* E.g., NH3 + NH4Cl

pH = pKa + log([A-]/[HA])

Henderson-Hasselbalch [HL] HL

Indicators

* Indicator (HIn) is itself a weak acid: HIn <-> H+ + In-
* Different colors for HIn and In- form
* Endpoint = color change midpoint of indicator range
* Choose indicator whose range includes equivalence point pH

[HL] HL EXTENSION

Buffer works best at pH = pKa. At the half-equivalence point in titration, pH = pK finding Ka experimentally.

[!] COMMON MISTAKE

For weak acids: never assume full dissociation. Use Ka expression. Common error: using [H+] = [HA]initial -- this gives pH that's too low.

9 REDOX & ELECTROCHEMISTRY

Oxidation States . Half Equations . Cells

Oxidation State Rules

Species	Oxidation State
Free element	0 (e.g., O2, Fe, Na)
Simple ion	Equals charge (e.g., Na+ = +1)
H in compounds	+1 (except metal hydrides: -1)
O in compounds	-2 (except peroxides: -1, OF2: +2)
F always	-1
Sum in neutral molecule	= 0
Sum in polyatomic ion	= charge of ion

OIL RIG Mnemonic

* **O**xidation **I**s **L**oss (of electrons)
* **R**eduction **I**s **G**ain (of electrons)
* **Oxidizing agent:** gains electrons (is itself reduced)
* **Reducing agent:** loses electrons (is itself oxidized)

Balancing Half-Equations (Acidic)

* 1. Write unbalanced half-equation
* 2. Balance atoms other than O and H
* 3. Balance O by adding H2O
* 4. Balance H by adding H+
* 5. Balance charge by adding e-
* 6. Combine: multiply so e- cancel
* **In alkaline solution:** add OH- instead of H+

[!] COMMON MISTAKE

When balancing redox in acidic conditions, always add H2O to balance O, then H+ to balance H. Never add O2 or H2 directly.

Electrochemical Cells

* **Galvanic (voltaic) cell:** spontaneous reaction -> electrical energy
* **Electrolytic cell:** electrical energy -> non-spontaneous reaction

Galvanic Cell Conventions

* Anode (-): oxidation occurs; loses mass
* Cathode (+): reduction occurs; gains mass
* Salt bridge: maintains electrical neutrality
* Edegcell = Edegcathode - Edeganode (from data booklet)
* Edegcell > 0 -> spontaneous reaction

Standard Electrode Potentials [HL] HL

* Measured vs. standard hydrogen electrode (SHE, Edeg = 0V)
* More positive Edeg: better oxidizing agent (more easily reduced)
* More negative Edeg: better reducing agent (more easily oxidized)
* Edegcell = Edeg(more +ve) - Edeg(more -ve)

Electrolysis

* Anode (+): oxidation (anions discharge -> e.g., Cl- -> Cl2)
* Cathode (-): reduction (cations discharge -> e.g., Cu2+ -> Cu)
* **Faraday's law:** n(e-) = Q/F; F = 96 485 C mol-1
* Competing discharges: in dilute H2SO4, O2 forms at anode; in concentrated HCl, Cl2 forms

[>] EXAM INSIGHT

At cathode: most easily reduced (most +ve Edeg) discharges. At anode: most eas

In dilute solutions: water/OH- preferred at anode.

10

ORGANIC CHEMISTRY

Functional Groups . Reactions . Mechanisms

Functional Groups & Homologous Series

Class	Functional Group	Suffix/Prefix	Example	Notes
Alkane	C-C (no pi bonds)	-ane	Ethane C2H6	CnH2n+2
Alkene	C=C (double bond)	-ene	Ethene C2H4	CnH2n
Alkyne [HL]HL	C=C (triple bond)	-yne	Ethyne C2H2	CnH2n-2
Halogenoalkane	C-X (X=halogen)	halo-	Chloromethane CH3Cl	X = F, Cl, Br, I
Alcohol	-OH (hydroxyl)	-ol	Ethanol C2H5OH	Primary/sec/tert
Aldehyde	-CHO	-al	Ethanal CH3CHO	Reducing agent
Ketone	C=O (not at end)	-one	Propanone CH3COCH3	Not easily oxidized
Carboxylic acid	-COOH	-oic acid	Ethanoic acid CH3COOH	Weak acid
Ester	-COO-	-oate	Ethyl ethanoate	From acid + alcohol
Amine [HL]HL	-NH2	-amine	Methylamine CH3NH2	Weak base
Amide [HL]HL	-CONH2	-amide	Ethanamide CH3CONH2	In peptide bonds
Benzene [HL]HL	Aromatic ring	phenyl-	Benzene C6H6	Resonance stabilized

[TIP] EXAM TIP

Know the systematic naming rules: longest carbon chain = parent name. Number to give functional group lowest locant. Alphabetical order for substituents.

Key Reaction Types

Reaction	Reactant class	Reagent/Condition	Product	Mechanism
Combustion	All organics	O2, ignition	CO2 + H2O	--
Free radical substitution	Alkanes	X2, UV light	Halogenoalkane + HX	Free radical (HL)
Electrophilic addition	Alkenes	HX, X2, H2/Ni, H2O/H+	Halogenoalkane/alkane/alcohol	Electrophilic add. (HL)
Nucleophilic substitution	Halogenoalkanes	NaOH(aq), KCN, NH3	Alcohol, nitrile, amine	SN1 / SN2 (HL)
Elimination	Halogenoalkanes	NaOH/ethanol (hot)	Alkene + HX	E2 (HL)
Esterification	Carboxylic acid + alcohol	H2SO4 catalyst, heat	Ester + H2O	Nucleophilic (HL)
Saponification	Ester	NaOH(aq), reflux	Alcohol + carboxylate salt	--
Oxidation of alcohol	Primary alcohol	K2Cr2O7/H2SO4, distil	Aldehyde (reflux->acid)	--
Reduction [HL]HL	Carbonyl compounds	NaBH4, LiAlH4	Alcohol	Nucleophilic add. (HL)
Electrophilic subst. [HL]HL	Benzene	Br2/FeBr3; HNO3/H2SO4	Bromobenzene; nitrobenzene	SE Ar (HL)

[HL] HL Reaction Mechanisms

Free Radical Substitution (Alkanes + Cl2/Br2)

* **Initiation:** Cl2 -> 2Cl* (UV light breaks homolytically)
* **Propagation 1:** Cl* + CH4 -> CH3* + HCl
* **Propagation 2:** CH3* + Cl2 -> CH3Cl + Cl*
* **Termination:** two radicals combine -> stable molecule
* Produces mixture of products; selectivity is poor

Electrophilic Addition (Alkenes)

* E.g., CH2=CH2 + HBr -> CH3CH2Br
* **Step 1:** H+ adds to double bond -> carbocation intermediate
* **Step 2:** Br- (nucleophile) attacks carbocation -> product
* **Markovnikov's rule:** H adds to C with more H (more stable carbocation)
* Bromine water test: Br2 (orange) -> decolorized = unsaturation

Electrophilic Substitution (Benzene) [HL]HL

* E.g., Benzene + Br2/FeBr3 -> bromobenzene + HBr
* **Step 1:** FeBr3 activates Br2 -> Br+ (electrophile)
* **Step 2:** Br+ attacks ring -> arenium ion (sigma complex)
* **Step 3:** Loss of H+ restores aromaticity -> bromobenzene
* Substitution (not addition) preserves aromatic stability

Nucleophilic Substitution (Halogenoalkanes)

* **SN2 (primary):** one-step backside attack -> inversion of configuration
* Nu:- attacks same time as X leaves; rate = k[RX][Nu]
* **SN1 (tertiary):** two-step, carbocation intermediate -> racemization
* Carbocation forms first; rate = k[RX] only
* **Common nucleophiles:** OH-, CN-, NH3, RO-

Condensation Polymerization [HL]HL

* **Polyesters:** dicarboxylic acid + diol -> polyester + H2O
* E.g., Terylene: terephthalic acid + ethylene glycol
* **Polyamides:** diamine + diacid -> polyamide + H2O
* E.g., Nylon-6,6: hexanedioic acid + 1,6-diaminohexane
* **Hydrolysis:** reverse of condensation -- add water with acid/base

Amino Acids & Proteins [HL]HL

* Amino acid: -NH2 and -COOH on same carbon (a-carbon)
* Peptide bond: -CO-NH- formed by condensation
* At pH 7: zwitterion form (NH3+ and COO-)
* Isoelectric point (pI): pH at which molecule has no net charge

[HL] HL EXTENSION

For paper 2: always show curly arrows (electron movement), lone pairs on nucleo charges (d+ / d-) on electrophiles for full marks.

[>] EXAM INSIGHT

Nucleophile = electron-rich, attacks d+ center. Electrophile = electron-poor, attacks electron-rich site. Learn to identify these in any mechanism.

Organic Reaction Pathway Map

Starting Material	Reaction	Reagents & Conditions	Product
Alkane	-> Halogenoalkane	X2, UV light (photochemical)	Halogenoalkane
Alkene	-> Halogenoalkane	HX (g), room temp	Halogenoalkane
Alkene	-> Dihaloalkane	X2 (l or aq), room temp	Dihalogenoalkane
Alkene	-> Alkane	H2, Ni catalyst, 150 degC	Alkane (hydrogenation)
Alkene	-> Alcohol	Steam + H3PO4 catalyst, 300 degC/70atm	Alcohol
Halogenoalkane	-> Alcohol	NaOH(aq), warm/reflux	Alcohol (SN)
Halogenoalkane	-> Nitrile [HL]HL	KCN, ethanol, reflux	Nitrile (SN)
Halogenoalkane	-> Amine [HL]HL	NH3 (excess), ethanol, heat	Amine (SN)
Halogenoalkane	-> Alkene	NaOH/ethanol, hot/reflux	Alkene (elimination)
Primary alcohol	-> Aldehyde	K2Cr2O7/H+, distil	Aldehyde
Primary alcohol	-> Carboxylic acid	K2Cr2O7/H+, reflux	Carboxylic acid
Secondary alcohol	-> Ketone	K2Cr2O7/H+, reflux	Ketone
Carboxylic acid	-> Ester	Alcohol + H2SO4 catalyst, reflux	Ester + H2O
Ester	-> Alcohol + acid	NaOH(aq), reflux (hydrolysis)	Alcohol + carboxylate
Aldehyde	-> Alcohol [HL]HL	NaBH4 in ethanol/water	Primary alcohol
Ketone	-> Alcohol [HL]HL	NaBH4 in ethanol/water	Secondary alcohol
Alcohol	-> Halogenoalkane	HX (conc), or PCl3/PCl5	Halogenoalkane
Alcohol	-> Alkene	Conc H2SO4, 170 degC (dehydration)	Alkene + H2O

[>] EXAM INSIGHT
Learn to write full reagents + conditions for every interconversion. Examiners award marks specifically for conditions (temperature, catalyst, solvent). "NaOH" alone is not enough -- specify "NaOH(aq)" vs "NaOH/ethanol".

EXAM MASTERY

[!] Top 15 Common Mistakes in IB Chemistry

1	Stoichiometry	Forgetting to convert cm3 -> dm3 (/1000) or degC -> K (+273) in PV=nRT
2	Stoichiometry	Using moles of the wrong reactant for theoretical yield -- always use the limiting reagent
3	Atomic Structure	Writing electron configs for ions incorrectly -- remove e- from highest n shell first (Fe2+: not [Ar]3d4, correct is [Ar]3d6)
4	Periodicity	Explaining IE trends without mentioning all three factors: nuclear charge, shielding, AND distance
5	Bonding	Confusing polar bonds with polar molecules -- a molecule can have polar bonds and still be nonpolar (e.g., CO2, CCl4)
6	Bonding	Forgetting lone pairs reduce bond angles in VSEPR -- must state "lone pair repels more strongly than bonding pair"
7	Energetics	Sign error in calorimetry: if T rises, reaction is exothermic so DH = NEGATIVE
8	Kinetics	Stating reaction order from stoichiometric coefficients -- orders are ONLY determined experimentally
9	Equilibrium	Saying catalyst shifts equilibrium -- it does NOT shift the position, only speeds up reaching it
10	Equilibrium	Changing Kc with pressure or concentration -- Kc only changes with TEMPERATURE
11	Acids & Bases	Applying strong acid formula to weak acids -- must use Ka expression and ICE table for weak acids
12	Acids & Bases	Forgetting that higher Ka = stronger acid, lower pKa = stronger acid (inverse log)
13	Redox	Confusing anode and cathode in electrolytic vs galvanic cells -- in BOTH: anode = oxidation, cathode = reduction
14	Organic	Writing "NaOH" without specifying conditions -- NaOH(aq) gives substitution; NaOH/ethanol gives elimination
15	Organic	Omitting curly arrows on nucleophile/electrophile in mechanisms -- examiners require arrow tails on lone pairs or bonds, heads on atoms

[>] IB Chemistry Exam Strategy

Paper 1 (MCQ -- 1 hour)

* 30 (SL) / 40 (HL) questions, no partial credit
* Use process of elimination -- rule out 2 wrong answers
* Always check units in calculation questions
* Mark difficult questions to return to -- don't get stuck
* Spend ~1 min per question; leave 10 min for review

Paper 2 (Short & Long Answer)

* **Read the command term:** "State" vs "Explain" vs "Deduce" require different depth
* "State": 1 mark, no justification needed
* "Explain": must include why/reasoning for full marks
* "Deduce": draw conclusion from data provided
* "Describe": list observations or features
* "Outline": brief explanation, no detail required

Extended Response (5-8 marks)

* Structure your answer with clear steps
* Show all working in calculations -- partial marks available
* Use correct scientific terminology throughout
* Include units at every step in calculations
* Sketch graphs: label axes, show correct shape and intercepts

Paper 3 (Options / HL Only)

* Choose option based on your strengths (if applicable)
* First question usually tests data analysis from experiment
* Graph questions: read scale carefully, identify anomalous points
* Error analysis: always quote uncertainty if given in data

Time Management

Paper	Duration	Time per mark
Paper 1 (SL)	45 min	1.5 min / mark
Paper 1 (HL)	60 min	1.5 min / mark
Paper 2 (SL)	75 min	1.25 min / mark
Paper 2 (HL)	135 min	1.25 min / mark
Paper 3 (HL)	60 min	1.25 min / mark

Calculation Questions -- Checklist

* [v] Convert all units before substituting
* [v] Write the formula clearly
* [v] Substitute with units
* [v] Round to correct sig figs (use data booklet values)
* [v] Include units in the final answer
* [v] Check sign (positive/negative) makes chemical sense

[TIP] EXAM TIP

The IB data booklet is your best friend. Practice using it before exams -- know wh
constant, and table is located.

[>] EXAM INSIGHT

Command terms determine the depth of your answer. "Suggest" means educated guess is acceptable.
"Calculate" means show full working. Never confuse them.

[F] Master Formula Reference

n = m / M
Moles from mass

n = cV
Moles from concentration

PV = nRT
Ideal gas law (R=8.314)

% yield = actual/theoreticalx100
Percentage yield

pH = -log[H+]
pH from concentration

Kw = [H+][OH-] = 10-14
Water equilibrium at 25 degC

Ka = [H+][A-]/[HA]
Weak acid constant

q = mcDT
Heat energy (c=4.18 for water)

DH = -q/n
Molar enthalpy

degrxn = SDHfdeg(prod)-SDHfdeg(rea
Hess's Law (formation)

Hdegrxn = SBE(broken)-SBE(formed
Bond energy method

Rate = k[A]m[B]n
Rate equation (HL)

k = Ae^(-Ea/RT)
Arrhenius equation (HL)

Kc = [prod]c/[react]a
Equilibrium constant

DG = DH - TDS
Gibbs free energy (HL)

Edegcell = Edegcathode-Edeganode
Cell potential (HL)

n(e-) = Q/F; F=96485 C/mol
Faraday's law (HL)

economy = M(desired)/SM(products)
Atom economy

[v] Final Revision Checklist -- Last 48 Hours

STOICHIOMETRY

- [ ] Can I calculate moles from mass, volume, and concentration?
- [ ] Can I identify the limiting reagent and calculate % yield?
- [ ] Can I use PV=nRT with correct unit conversions?

ATOMIC STRUCTURE

- [ ] Can I write electron configurations including ions and transition metals?
- [ ] Can I explain IE trends and exceptions (B<Be, O<N)?

PERIODICITY

- [ ] Can I explain trends in atomic radius, EN, and IE across period and down group?
- [ ] Do I know the reactions of Period 3 oxides and chlorides with water?

BONDING

- [ ] Can I draw Lewis structures including expanded octets (HL)?
- [ ] Can I determine VSEPR shapes and bond angles from formula?
- [ ] Can I identify and rank IMFs and explain their effect on physical properties?

ENERGETICS

- [ ] Can I calculate DH from calorimetry data (q=mcDT)?
- [ ] Can I use Hess's Law cycle and bond energy methods?
- [ ] Do I know Born-Haber cycles and DG = DH - TDS (HL)?

KINETICS

- [ ] Can I determine rate order from experimental data?
- [ ] Can I sketch and interpret Maxwell-Boltzmann curves?
- [ ] Do I know the Arrhenius equation and how to find Ea (HL)?

EQUILIBRIUM

- [ ] Can I write Kc expressions correctly?
- [ ] Can I apply Le Chatelier's principle to all changes?
- [ ] Do I know why catalyst doesn't shift equilibrium position?

ACIDS & BASES

- [ ] Can I calculate pH for strong and weak acids/bases?
- [ ] Can I describe and sketch all four titration curve shapes?
- [ ] Do I understand buffer action and Henderson-Hasselbalch (HL)?

REDOX

- [ ] Can I assign oxidation states in any compound?
- [ ] Can I balance half-equations in acidic and basic conditions?
- [ ] Can I explain anode vs cathode in both cell types?

ORGANIC

- [ ] Can I name and draw all functional groups?
- [ ] Do I know reagents + conditions for every reaction type?
- [ ] Can I draw complete mechanisms with curly arrows (HL)?

[TIP] EXAM TIP

You've got this. Every formula in this sheet, every reaction pathway, every exam tip -- it's all here.

Trust your preparation, read each question carefully, and show your working. Good luck! :)